Bases, Dimension, and Rank

Linear Algebra · Lecture 4

Last time (Lecture 3): subspaces, spans, linear independence and bases. Here $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m\times n}$ has the columns a_j; e_j is column j of I; row_i, col_j are row i and column j.

1. Bases and dimension of subspaces of $\mathbb{R}^n$

Theorem 1. Let R be an RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows. The following are equivalent: (i) the columns $a_1, \dots, a_n$ of A are linearly independent; (ii) $Ax = 0$ only for $x = 0$; (iii) $r = n$, that is, every column of R is a pivot column. In particular, if $n > m$, then any n vectors of $\mathbb{R}^m$ are linearly dependent.

Proof.

Lecture 3, Proposition 11 (for $V = [v_1 \ \cdots \ v_k]$: $v_1, \dots, v_k$ independent $\iff$ $Vc = 0$ only for $c = 0$ $\iff$ every column of an RREF of V is a pivot column; more than m vectors of $\mathbb{R}^m$ are dependent), with $V = A$; Lecture 2, Definition 4(a),(d):

(i) $\iff$ (ii) $\iff$ every column of R is a pivot column; $\quad n > m \implies$ dependent
each nonzero row of R has one pivot, in different columns $\implies$ r pivot columns
every column of R is a pivot column $\iff$ $r = n$

□

Theorem 2. Let R be an RREF of $A \in \mathbb{R}^{m\times n}$, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and entries R_{ij}.

(a) For $x \in \mathbb{R}^n$: $x_1a_1 + \cdots + x_na_n = 0 \iff x_1 \,\text{col}_1(R) + \cdots + x_n \,\text{col}_n(R) = 0$.

(b) If $r \geqslant 1$, the **pivot columns** $a_{k_1}, \dots, a_{k_r}$ of A form a basis of $\text{span}\{a_1, \dots, a_n\}$, and

$$a_j = R_{1j}a_{k_1} + R_{2j}a_{k_2} + \cdots + R_{rj}a_{k_r} \qquad (j = 1, \dots, n)$$

If $r = 0$, then $A = 0$ and $\text{span}\{a_1, \dots, a_n\} = \{0\}$.

In particular, every finite list that spans a subspace W contains a basis of W.

Proof.

(a) Lecture 1, Theorem 4(b); Lecture 2, Theorem 3(b) for $[A \mid 0] \to [R \mid 0]$ (row operations keep the solutions):

$$x_1a_1 + \cdots + x_na_n = Ax, \quad x_1 \,\text{col}_1(R) + \cdots + x_n \,\text{col}_n(R) = Rx; \qquad Ax = 0 \iff Rx = 0$$

(b) $r = 0$: $R = 0$, and $R = PA$ with P invertible (Lecture 2, Theorem 14(c)), so $A = P^{-1}R = 0$. Let $r \geqslant 1$.

Relations: $\text{col}_{k_i}(R) = e_i$, and rows $r+1, \dots, m$ of R are 0:

$\text{col}_j(R) = \sum_{i=1}^r R_{ij}e_i = \sum_{i=1}^r R_{ij} \,\text{col}_{k_i}(R)$ (every j)
$j \notin \{k_1, \dots, k_r\}$: $x_j = 1$, $x_{k_i} = -R_{ij}$, other $x_l = 0 \implies Rx = \text{col}_j(R) - \sum_i R_{ij} \,\text{col}_{k_i}(R) = 0$
$\implies Ax = a_j - \sum_i R_{ij}a_{k_i} = 0$ ((a))
$j = k_l$: $R_{ik_l} = 1$ $(i = l)$, 0 $(i \neq l) \implies$ the formula reads $a_{k_l} = a_{k_l}$

Basis (Lecture 3, Definition 14: independent vectors of W that span W): $U = \text{span}\{a_{k_1}, \dots, a_{k_r}\}$; a span is the smallest subspace containing its vectors (Lecture 3, Theorem 7):

$a_1, \dots, a_n \in U \implies \text{span}\{a_1, \dots, a_n\} \subseteq U \subseteq \text{span}\{a_1, \dots, a_n\}$ (Lecture 3, Theorem 7)
$c_1a_{k_1} + \cdots + c_ra_{k_r} = 0 \implies c_1e_1 + \cdots + c_re_r = 0 \implies c_1 = \cdots = c_r = 0$ ((a))

Last sentence: write the spanning vectors as the columns of a matrix and apply (b); if $r = 0$, then $W = \{0\}$, and the empty list is a basis of W. □

Pivot columns of A. Let R be an RREF of A, with pivot columns $k_1 < \cdots < k_r$.

- By Theorem 2(a) (take $x_j = -1$, $x_l = 0$ for $l > j$), a_j is a linear combination of $a_1, \ldots, a_{j-1} \iff$ $\mathrm{col}_j(R)$ is a linear combination of $\mathrm{col}_1(R), \ldots, \mathrm{col}_{j-1}(R)$.
- In R this happens exactly when j is not a pivot column: $\mathrm{col}_{k_i}(R) = e_i$, while the columns before k_i are 0 in rows $i, \ldots, m$ (Lecture 2, Definition 4(a),(c),(d)); and a column $j \notin \{k_1, \ldots, k_r\}$ is $\sum_{k_i<j} R_{ij}\,\mathrm{col}_{k_i}(R)$ (proof of Theorem 2(b), first line; row i of R is 0 before column k_i).
- So the pivot columns of A are the columns that are not combinations of the earlier ones: they, and their number r, do not depend on the row operations used.

Remark 3. Let H be an echelon form of A (Lecture 2, Definition 4: each nonzero row begins with a pivot 1, zero rows are last, the pivot columns increase), with pivots in columns $k_1 < \cdots < k_r$. Then the pivot columns of A are $a_{k_1}, \ldots, a_{k_r}$, and r is the number of nonzero rows of H. So for Theorem 1 one may stop at an echelon form: (iii) says that every column of H contains a pivot. The same holds for a matrix reached from A whose zero rows are last and whose first nonzero entries, not yet scaled to 1, lie in increasing columns.

Proof.
- Backward elimination adds multiples of a row i, whose pivot is in column k_i, to rows $l < i$, whose pivot is in a column $k_l < k_i$. Row i is 0 before column k_i, so row l changes only in columns $\geqslant k_i$ and keeps its pivot in column k_l. Row i is also 0 in the pivot columns $k_{i+1}, \ldots, k_r$ already cleared, so they stay $e_{i+1}, \ldots, e_r$; the zero rows of H do not change. So the result is in RREF (Lecture 2, Definition 5).
- So backward elimination takes H to an RREF of A with the same pivot columns $k_1, \ldots, k_r$ as H; by the paragraph above, these are the pivot columns of A.
- Scaling a row by $c \neq 0$ does not move its first nonzero entry. □

Theorem 4. Every subspace W of $\mathbb{R}^n$ has a basis, and any two bases of W have the same number of elements, the **dimension** $\dim W$ of W. $\dim \mathbb{R}^n = n$ (standard basis), $\dim\{0\} = 0$ (empty list), and $\dim W = 0$ only for $W = \{0\}$. If W is spanned by m vectors, then any $k > m$ vectors of W are linearly dependent.

Proof.

$W = \mathrm{span}\{w_1, \ldots, w_m\}$, $v_1, \ldots, v_k \in W$, $k > m$ (if $m = 0$: $W = \{0\}$, and $k \geqslant 1$ vectors of $\{0\}$ are dependent, Lecture 3, Proposition 12(a): a list containing 0 is dependent); $m \geqslant 1$:

$$
\begin{aligned}
&v_j = a_{1j}w_1 + \cdots + a_{mj}w_m \ (j = 1, \ldots, k), \quad A = (a_{ij}) \in \mathbb{R}^{m\times k} \\
&k > m \implies Ax = 0 \text{ for some } x \neq 0 \qquad \text{(Theorem 1)} \\
&x_1v_1 + \cdots + x_kv_k = \textstyle\sum_{j=1}^{k} x_j \sum_{i=1}^{m} a_{ij}w_i = \sum_{i=1}^{m} (Ax)_i w_i = 0 \\
&\text{bases } v_1, \ldots, v_k \text{ and } w_1, \ldots, w_m \text{ of } W \implies k \leqslant m \text{ and } m \leqslant k
\end{aligned}
$$

- Existence: start with the empty list and, as long as the vectors chosen do not span W, adjoin a vector $v \in W$ outside their span. In a relation $c_1v_1 + \cdots + c_kv_k + cv = 0$, $c \neq 0$ would put v in the span of $v_1, \ldots, v_k$; so $c = 0$, and then every $c_i = 0$: the list stays independent.
- More than n vectors of $\mathbb{R}^n$ are dependent (Theorem 1), so this stops after at most n steps, at a basis of W.
- A basis with no vectors is the empty list, which spans $\{0\}$ (Lecture 3, Definition 6: $\mathrm{span}\,\varnothing = \{0\}$). □

Theorem 5. Let W be a subspace of $\mathbb{R}^n$ with $\dim W = d$.

(a) If U is a subspace of W, then $\dim U \leqslant d$, and $\dim U = d$ only for $U = W$. In particular $d \leqslant n$, and $d = n$ only for $W = \mathbb{R}^n$.

(b) Every linearly independent list of vectors of W can be extended to a basis of W.

(c) More than d vectors of W are linearly dependent; fewer than d vectors do not span W.

(d) d vectors of W are linearly independent $\iff$ they span W $\iff$ they form a basis of W.

Proof. $d = 0$: $W = \{0\}$ (Theorem 4).

- Every subspace U of W is $\{0\} = W$; a nonempty list of vectors of W contains 0, so it is dependent (Lecture 3, Proposition 12(a)); the empty list, the only independent one, is a basis of W (Lecture 3, Definition 14: the empty list is a basis of $\{0\}$). So (a)–(c) hold.
- (d): the only list of 0 vectors is the empty list; it is independent, spans $\{0\} = W$ and is a basis of W.

Let $d \geqslant 1$, and let $w_1, \dots, w_d$ be a basis of W (Theorem 4). We prove (c) first.

(c) $w_1, \dots, w_d$ span W; Theorem 4, last sentence:

$k > d$ vectors of W are dependent $(m = d)$

$k < d$ vectors spanning $W \implies w_1, \dots, w_d$ $(d > k)$ dependent ✘ $(m = k)$

(b)

- Let $v_1, \dots, v_k \in W$ be independent. If they span W, they form a basis of W.
- Otherwise take $v \in W$ outside $\operatorname{span}\{v_1, \dots, v_k\}$. Then $v_1, \dots, v_k, v$ are independent: in $c_1v_1 + \dots + c_kv_k + cv = 0$, $c \neq 0$ would put v in the span, so $c = 0$, and then every $c_i = 0$.
- Repeat. Each list obtained is independent in W, so it has at most d vectors by (c); the process stops, at a basis of W that contains $v_1, \dots, v_k$.

(a) A basis of U (Theorem 4) consists of $\dim U$ independent vectors of W:

$\dim U \leqslant d$ ((c))

$U \neq W$, $v \in W \setminus U \implies$ a basis of U, and v: $\dim U + 1$ independent vectors of W (as in (b))

$\implies \dim U + 1 \leqslant d$; so $\dim U = d$ only for $U = W$ ((c))

$W \subseteq \mathbb{R}^n$, $\dim \mathbb{R}^n = n \implies d \leqslant n$, $d = n$ only for $W = \mathbb{R}^n$ (Theorem 4)

(d) A basis of W is independent and spans W (Lecture 3, Definition 14). Let $v_1, \dots, v_d \in W$.

- If they are independent and $w \in W$: by (c), $x_0w + x_1v_1 + \dots + x_dv_d = 0$ with not all $x_i = 0$; $x_0 \neq 0$, as $v_1, \dots, v_d$ are independent; so $w = -(x_1/x_0)v_1 - \dots - (x_d/x_0)v_d$. They span W: a basis.
- If they span W: by Theorem 2, some of them form a basis of W, which has d vectors (Theorem 4); so it is all of $v_1, \dots, v_d$. □

Transposition carries a subspace of $\mathbb{R}^{1\times n}$ onto a subspace of $\mathbb{R}^n$ and preserves linear combinations, hence independence, spanning and bases; so Theorems 4 and 5 also hold for subspaces of $\mathbb{R}^{1\times n}$, such as the row space $\mathcal{R}(A)$ below.

Example 6. Let $W = \operatorname{span}\{u_1, \dots, u_5\} \subseteq \mathbb{R}^4$, where $u_1 = (1,2,1,1)'$, $u_2 = (1,3,2,1)'$, $u_3 = (1,1,0,1)'$, $u_4 = (1,2,2,3)'$, $u_5 = (1,3,1,-1)'$. (a) Are $u_1, \dots, u_5$ linearly independent? (b) Find a basis of W among $u_1, \dots, u_5$, and $\dim W$. (c) Express the remaining vectors in this basis.

Solution.

Row-reduce $A = [u_1 \ \cdots \ u_5]$:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 1 \\ 1 & 1 & 1 & 3 & -1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1,\ R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 2 & -2 \end{pmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 & -2 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\xrightarrow{R_1 \leftarrow R_1 - R_3} \begin{pmatrix} 1 & 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = R$$

$r = 3$, pivot columns $k_1 = 1$, $k_2 = 2$, $k_3 = 4$:

(a) No: $r = 3 < 5$; indeed any five vectors of $\mathbb{R}^4$ are dependent (Theorem 1)

(b) u_1, u_2, u_4 form a basis of W, $\dim W = 3$; $3 < 4 \implies W \neq \mathbb{R}^4$ (Theorems 2(b), 5(a))

(c) columns 3, 5 of R: $u_3 = 2u_1 - u_2$, $\quad u_5 = u_1 + u_2 - u_4$ (Theorem 2(b))

Check: $2(1,2,1,1)' - (1,3,2,1)' = (1,1,0,1)' = u_3$ and $(1,2,1,1)' + (1,3,2,1)' - (1,2,2,3)' = (1,3,1,-1)' = u_5$.

2. The four subspaces of a matrix

Definition 7. For $A \in \mathbb{R}^{m \times n}$, the **column space**, **row space**, **null space** and **left null space** of A are

$$\mathcal{C}(A) = \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, \qquad \mathcal{R}(A) = \{y'A : y \in \mathbb{R}^m\} \subseteq \mathbb{R}^{1 \times n}$$
$$\mathcal{N}(A) = \{x \in \mathbb{R}^n : Ax = 0\}, \qquad \mathcal{N}(A') = \{y \in \mathbb{R}^m : A'y = 0\}$$

The columns of A span $\mathcal{C}(A)$, the rows of A span $\mathcal{R}(A)$, and $y \in \mathcal{N}(A') \iff y'A = 0$ (transpose). All four are subspaces (Lecture 3, Theorem 7: a span is a subspace; Lecture 3, Example 5(b): $\{x : Ax = 0\}$ is a subspace).

Let R be an RREF of A, with r nonzero rows, pivot columns $k_1 < \cdots < k_r$ and **free columns** $f_1 < \cdots < f_{n-r}$ (the free unknowns of $Ax = 0$ are $x_{f_1}, \dots, x_{f_{n-r}}$). For $j = 1, \dots, n-r$, the **special solution** $s_j \in \mathbb{R}^n$ is the solution of $Ax = 0$ with $x_{f_j} = 1$ and $x_{f_l} = 0$ $(l \neq j)$. By Lecture 2, Proposition 7(b) (all solutions of $Ax = 0$: free unknowns arbitrary, $x_{k_i} = -\sum_{j \text{ free}} R_{ij} x_j$), its entries are

$$(s_j)_{f_l} = \delta_{jl} \ (1 \leqslant l \leqslant n-r), \qquad (s_j)_{k_i} = -R_{if_j} \ (1 \leqslant i \leqslant r)$$

where $\delta_{jl} = 1$ if $j = l$ and 0 otherwise.

Theorem 8. Let $A \in \mathbb{R}^{m\times n}$, and let row operations take $[A \mid I_m]$ to $[R \mid P]$ with R in RREF (an RREF of A), with nonzero rows $\rho'_1, \dots, \rho'_r$ ($\rho_i \in \mathbb{R}^n$), pivot columns $k_1 < \cdots < k_r$ and free columns $f_1 < \cdots < f_{n-r}$. Then $P \in \mathbb{R}^{m\times m}$ is invertible, $PA = R$, and:

(a) if $r \geqslant 1$, the pivot columns $a_{k_1}, \dots, a_{k_r}$ of A form a basis of $\mathcal{C}(A)$;

(b) $\mathcal{R}(R) = \mathcal{R}(A)$, and if $r \geqslant 1$, the nonzero rows $\rho'_1, \dots, \rho'_r$ of R form a basis of $\mathcal{R}(A)$;

(c) the special solutions $s_1, \dots, s_{n-r}$ form a basis of $\mathcal{N}(A)$ (if $r = n$: $\mathcal{N}(A) = \{0\}$, empty basis);

(d) $\mathrm{row}_{r+1}(P)', \dots, \mathrm{row}_m(P)'$ form a basis of $\mathcal{N}(A')$ (if $r = m$: $\mathcal{N}(A') = \{0\}$, empty basis).

Hence $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, $\dim \mathcal{N}(A) = n - r$, $\dim \mathcal{N}(A') = m - r$ (if $r = 0$, then $A = 0$ and $\mathcal{C}(A) = \mathcal{R}(A) = \{0\}$).

Proof.

P and $PA = R$: the row operations multiply on the left by an invertible E (Lecture 2, Theorem 14(c)); column by column (Lecture 1, Theorem 4(a)):

$$E[A \mid I_m] = [EA \mid E] = [R \mid P] \implies P = E \text{ is invertible, } PA = R$$

(a) $\mathcal{C}(A) = \mathrm{span}\{a_1, \dots, a_n\}$; Theorem 2(b).

(b) $P = (p_{ij})$; Lecture 1, Theorem 4(c); Lecture 3, Theorem 7: a span is the smallest subspace containing its vectors:

$$\begin{aligned}
&\mathrm{row}_i(R) = \mathrm{row}_i(P)A = p_{i1}\,\mathrm{row}_1(A) + \cdots + p_{im}\,\mathrm{row}_m(A) \in \mathcal{R}(A) \implies \mathcal{R}(R) \subseteq \mathcal{R}(A)\\
&A = P^{-1}R \implies \mathcal{R}(A) \subseteq \mathcal{R}(R) \qquad \text{so } \mathcal{R}(R) = \mathcal{R}(A)\\
&\text{rows } r+1, \dots, m \text{ of } R \text{ are } 0 \implies \rho'_1, \dots, \rho'_r \text{ span } \mathcal{R}(R) = \mathcal{R}(A)\\
&\mathrm{col}_{k_i}(R) = e_i \implies \text{entry } k_i \text{ of } c_1\rho'_1 + \cdots + c_r\rho'_r \text{ is } c_i \implies \text{independent}
\end{aligned}$$

(c) $Ax = 0 \iff Rx = 0$ (Lecture 2, Theorem 3(b)); pivot unknowns from free ones (Lecture 2, Proposition 7(b)); $\mathcal{N}(A)$ is closed under linear combinations (Lecture 3, Theorem 4); $t \in \mathcal{N}(A)$, $\tilde{t} = t_{f_1}s_1 + \cdots + t_{f_{n-r}}s_{n-r}$:

$$\begin{aligned}
&\text{entry } f_l \text{ of } c_1s_1 + \cdots + c_{n-r}s_{n-r} \text{ is } \textstyle\sum_j c_j\delta_{jl} = c_l \implies \text{independent}\\
&\tilde{t} \in \mathcal{N}(A) \text{ (a subspace)}, \quad \tilde{t}_{f_l} = t_{f_l} \ (l = 1, \dots, n-r)\\
&x_{k_i} = -\textstyle\sum_l R_{if_l}x_{f_l} \text{ for every solution} \implies t = \tilde{t} \in \mathrm{span}\{s_1, \dots, s_{n-r}\}\\
&r = n: \text{ no free unknowns, so } x = 0 \text{ is the only solution}
\end{aligned}$$

(d) Lecture 1, Theorems 3(a), 4(c), 8; Lecture 3, Proposition 12(b): sublists of independent lists are independent:

$$\begin{aligned}
&i > r: \ \mathrm{row}_i(P)A = \mathrm{row}_i(PA) = \mathrm{row}_i(R) = 0 \implies A'\,\mathrm{row}_i(P)' = (\mathrm{row}_i(P)A)' = 0\\
&c'P = c_1\,\mathrm{row}_1(P) + \cdots + c_m\,\mathrm{row}_m(P) = 0 \implies c' = c'PP^{-1} = 0\\
&\quad\implies \text{the rows of } P \text{ are independent, and so are rows } r+1, \dots, m\\
&A'y = 0, \ z' = y'P^{-1}: \quad y' = z'P, \quad 0 = y'A = z'PA = z'R = z_1\rho'_1 + \cdots + z_r\rho'_r\\
&\quad\implies z_1 = \cdots = z_r = 0 \implies y' = \textstyle\sum_{i>r} z_i\,\mathrm{row}_i(P) \qquad ((b))
\end{aligned}$$

□

Example 9. Find bases of the four subspaces of $A = \left(\begin{smallmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{smallmatrix}\right)$.

Solution.

Row-reduce $[A \mid I_3]$:

$$\left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 5 & 0 & 1 & 0 \\ 2 & 5 & 1 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & -2 & 0 & 1 \end{array}\right)$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|ccc} 1 & 0 & 3 & -1 & 3 & -2 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array}\right) = [R \mid P]$$

$r = 2$, pivot columns $1, 2$, free unknowns x_3, x_4 (already visible in the echelon form after the second arrow; Remark 3).

$\mathcal{C}(A)$, $\mathcal{R}(A)$ (Theorem 8(a),(b)); columns 3, 4 of R (Theorem 2(b)):

$$\mathcal{C}(A): \ a_1 = (1,1,2)', \ a_2 = (2,3,5)'; \quad a_3 = 3a_1 - a_2, \quad a_4 = -a_1 + 2a_2; \quad \dim \mathcal{C}(A) = 2$$
$$\mathcal{R}(A): \ \rho_1' = (1,0,3,-1), \ \rho_2' = (0,1,-1,2); \quad \dim \mathcal{R}(A) = 2$$

Check: $3(1,1,2)' - (2,3,5)' = (1,0,1)' = a_3$ and $-(1,1,2)' + 2(2,3,5)' = (3,5,8)' = a_4$.

$\mathcal{N}(A)$, $\mathcal{N}(A')$ (Theorem 8(c),(d)):

$$Rx = 0: \ x_1 + 3x_3 - x_4 = 0, \ x_2 - x_3 + 2x_4 = 0$$
$$\mathcal{N}(A): \ s_1 = (-3,1,1,0)' \ (x_3 = 1, \ x_4 = 0), \quad s_2 = (1,-2,0,1)' \ (x_3 = 0, \ x_4 = 1)$$
$$\dim \mathcal{N}(A) = 2 = 4 - 2, \quad \dim \mathcal{N}(A') = 1 = 3 - 2$$
$$\mathcal{N}(A'): \ y = \operatorname{row}_3(P)' = (-1,-1,1)'$$
$$y'A = -\operatorname{row}_1(A) - \operatorname{row}_2(A) + \operatorname{row}_3(A) = 0: \quad \operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A)$$

Warning: every column of R has third entry 0, so every vector of $\mathcal{C}(R)$ does; but a_1 has third entry 2. So $\mathcal{C}(R) \neq \mathcal{C}(A)$: row operations keep $\mathcal{R}(A)$ and $\mathcal{N}(A)$ (Theorem 8(b); Lecture 2, Theorem 3(b)), not $\mathcal{C}(A)$, and a basis of $\mathcal{C}(A)$ consists of pivot columns of A itself, not of R.

3. Rank

Definition 10. By Theorem 8, $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, the number of pivots of an RREF of A (also when $r = 0$): the **rank** $\operatorname{rank} A$ of A; by Remark 3, it is the number of nonzero rows of any echelon form of A. $A \in \mathbb{R}^{m \times n}$ has **full column rank** if $\operatorname{rank} A = n$, **full row rank** if $\operatorname{rank} A = m$. $\dim \mathcal{N}(A)$ is the **nullity** of A.

Theorem 11. For every $A \in \mathbb{R}^{m \times n}$: $\operatorname{rank} A + \dim \mathcal{N}(A) = n$ and $\operatorname{rank} A + \dim \mathcal{N}(A') = m$.

Proof.

R as in Theorem 8, $\operatorname{rank} A = r$:

$$n = r \text{ (pivot columns)} + (n - r) \text{ (free columns)}, \quad \dim \mathcal{N}(A) = n - r \qquad \text{(Theorem 8(c))}$$
$$\dim \mathcal{N}(A') = m - r, \text{ the number of zero rows of } R \qquad \text{(Theorem 8(d))}$$

□

Theorem 12. Let $A \in \mathbb{R}^{m\times n}$.

(a) $\operatorname{rank} A' = \operatorname{rank} A$, and $\operatorname{rank} A \leqslant \min\{m, n\}$.

(b) A has full column rank $\iff$ its columns are linearly independent $\iff$ $\mathcal{N}(A) = \{0\}$.

(c) A has full row rank $\iff$ its rows are linearly independent $\iff$ $\mathcal{N}(A') = \{0\}$ $\iff$ $\mathcal{C}(A) = \mathbb{R}^m$.

(d) If $m = n$: A is invertible $\iff$ $\operatorname{rank} A = n$.

Proof. Let R be an RREF of A, with r nonzero rows and pivot columns $k_1 < \cdots < k_r$; $\operatorname{rank} A = r$ (Definition 10).

(a) $(y'A)' = A'y$ (Lecture 1, Theorem 8): transposing maps $\mathcal{R}(A)$ onto $\mathcal{C}(A')$, keeps linear combinations and is undone by transposing again:

$$\text{a basis of } \mathcal{R}(A) \text{ goes to a basis of } \mathcal{C}(A') \implies \operatorname{rank} A' = \dim \mathcal{C}(A') = \dim \mathcal{R}(A) = \operatorname{rank} A$$
$$r \leqslant m \ (R \text{ has } m \text{ rows}), \quad r \leqslant n \ (k_1 < \cdots < k_r \text{ lie in } \{1, \dots, n\})$$

(b) Full column rank means $r = n$ (Definition 10); so (b) is Theorem 1: (iii) $\iff$ (i) $\iff$ (ii).

(c) Lecture 1, Theorems 4(c), 8:

$$\operatorname{rank} A = m \iff \dim \mathcal{N}(A') = m - \operatorname{rank} A = 0 \iff \mathcal{N}(A') = \{0\} \qquad \text{(Theorems 11, 4)}$$
$$y'A = y_1 \operatorname{row}_1(A) + \cdots + y_m \operatorname{row}_m(A), \quad y'A = 0 \iff A'y = 0$$
$$\implies \big(\mathcal{N}(A') = \{0\} \iff \text{the rows of } A \text{ are independent}\big)$$
$$\operatorname{rank} A = \dim \mathcal{C}(A) = m \iff \mathcal{C}(A) = \mathbb{R}^m \qquad \text{(Theorems 5(a), 4)}$$

(d) Lecture 2, Theorem 15:

$$A \text{ invertible} \iff (Ax = 0 \implies x = 0) \iff \mathcal{N}(A) = \{0\} \iff \operatorname{rank} A = n \qquad \text{((b))}$$

□

Corollary 13. Let $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, and $[A \mid b] \in \mathbb{R}^{m\times(n+1)}$.

(a) $Ax = b$ is consistent $\iff$ $b \in \mathcal{C}(A)$ $\iff$ $\operatorname{rank}[A \mid b] = \operatorname{rank} A$.

(b) If $Ax = b$ is consistent, its solution is unique $\iff$ $\operatorname{rank} A = n$.

Proof.

(a) $Ax = x_1a_1 + \cdots + x_na_n$ (Lecture 1, Theorem 4(b)):

$$Ax = b \text{ has a solution} \iff b \in \mathcal{C}(A)$$
$$\mathcal{C}(A) \subseteq \mathcal{C}([A \mid b]) \quad \text{(coefficient 0 on } b\text{)}$$
$$b = Ac \implies x_1a_1 + \cdots + x_na_n + tb = A(x + tc) \in \mathcal{C}(A) \implies \mathcal{C}([A \mid b]) = \mathcal{C}(A)$$
$$\operatorname{rank}[A \mid b] = \operatorname{rank} A \implies \mathcal{C}(A) = \mathcal{C}([A \mid b]) \ni b \qquad \text{(Theorem 5(a))}$$

(b) Solutions $x_p + z$, $z \in \mathcal{N}(A)$ (Lecture 2, Theorem 22(a)):

$$\text{unique} \iff \mathcal{N}(A) = \{0\} \iff \operatorname{rank} A = n \qquad \text{(Theorem 12(b))}$$

□

Equations of $\mathcal{C}(A)$. Let row operations take $[A \mid I_m]$ to $[R \mid P]$ as in Theorem 8, $r = \operatorname{rank} A$, and $b \in \mathbb{R}^m$.

The same operations multiply on the left by E (Lecture 2, Theorem 14(c)); Lecture 1, Theorem 4(a); Corollary 13(a); Lecture 2, Proposition 7(a) (consistent $\iff z_i = 0$ for $i > r$) with $z = Pb$:

$$E[A \mid I_m] = [EA \mid E] = [R \mid P] \implies E = P, \quad E[A \mid b] = [R \mid Pb]$$
$$b \in \mathcal{C}(A) \iff Ax = b \text{ is consistent} \iff \operatorname{row}_i(P)\, b = 0 \quad (i = r+1, \dots, m)$$

No condition if $r = m$.

Example 14. For A of Example 9, when is $Ax = (1, t, 4)'$ consistent? Find all its solutions then.

Solution.

Example 9: $r = 2$, $\operatorname{row}_3(P) = (-1, -1, 1)$; the equations of $\mathcal{C}(A)$:

$$b \in \mathcal{C}(A) \iff -b_1 - b_2 + b_3 = 0 \quad (\text{the coefficients of } \operatorname{row}_3(A) = \operatorname{row}_1(A) + \operatorname{row}_2(A))$$
$$b = (1, t, 4)' : \ -1 - t + 4 = 0 \iff t = 3$$

$t = 3$, $b = (1, 3, 4)'$; $Ax = b \iff Rx = Pb$ (Lecture 2, Theorem 3(b)); all solutions $x_p + z$, $z \in \mathcal{N}(A)$ (Lecture 2, Theorem 22(a)), and Theorem 8(c):

$$Pb = (3 - 6, \ -1 + 3, \ -1 - 3 + 4)' = (-3, 2, 0)'$$
$$Rx = Pb : \ x_1 + 3x_3 - x_4 = -3, \ x_2 - x_3 + 2x_4 = 2; \quad x_3 = x_4 = 0 : \ x_p = (-3, 2, 0, 0)'$$
$$\text{all solutions: } x_p + c_1 s_1 + c_2 s_2 \ (c_1, c_2 \in \mathbb{R}); \quad \text{not unique: } \operatorname{rank} A = 2 < 4 \text{ (Corollary 13(b))}$$

Check: $Ax_p = -3a_1 + 2a_2 = (-3 + 4, \ -3 + 6, \ -6 + 10)' = (1, 3, 4)'$.

4. Rank equalities and inequalities

Theorem 15. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

(a) $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$, $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$, and $\operatorname{rank}(AB) \leqslant \min\{\operatorname{rank} A, \operatorname{rank} B\}$.

(b) If A has full column rank, then $\mathcal{N}(AB) = \mathcal{N}(B)$ and $\operatorname{rank}(AB) = \operatorname{rank} B$.

(c) If B has full row rank, then $\mathcal{C}(AB) = \mathcal{C}(A)$ and $\operatorname{rank}(AB) = \operatorname{rank} A$.

(d) $\operatorname{rank}(PAQ) = \operatorname{rank} A$ for all invertible $P \in \mathbb{R}^{m \times m}$ and $Q \in \mathbb{R}^{n \times n}$.

Proof.

(a) $(AB)' = B'A'$ (Lecture 1, Theorem 8); Theorem 12(a):

$$ABx = A(Bx) \in \mathcal{C}(A); \quad Bx = 0 \implies ABx = A0 = 0$$
$$\operatorname{rank}(AB) = \dim \mathcal{C}(AB) \leqslant \dim \mathcal{C}(A) = \operatorname{rank} A \qquad \text{(Theorem 5(a))}$$
$$\operatorname{rank}(AB) = \operatorname{rank}\big((AB)'\big) = \operatorname{rank}(B'A') \leqslant \operatorname{rank} B' = \operatorname{rank} B$$

(b) $\mathcal{N}(A) = \{0\}$ (Theorem 12(b)):

$$ABx = 0 \implies Bx \in \mathcal{N}(A) = \{0\} \implies x \in \mathcal{N}(B); \quad \text{with (a), } \mathcal{N}(AB) = \mathcal{N}(B)$$
$$\operatorname{rank}(AB) = p - \dim \mathcal{N}(AB) = p - \dim \mathcal{N}(B) = \operatorname{rank} B \qquad \text{(Theorem 11)}$$

(c) $\mathcal{C}(B) = \mathbb{R}^n$ (Theorem 12(c)):

$$x \in \mathbb{R}^n \implies x = Bw \implies Ax = (AB)w \in \mathcal{C}(AB) \implies \mathcal{C}(A) \subseteq \mathcal{C}(AB)$$
$$\text{with (a), } \mathcal{C}(AB) = \mathcal{C}(A), \text{ so } \operatorname{rank}(AB) = \operatorname{rank} A$$

(d) Theorem 12(b),(d):

$Px = 0 \implies x = P^{-1}Px = 0$: P has full column rank (Lecture 1, Theorem 3(a))
$\operatorname{rank} Q = n$: Q has full row rank
$\operatorname{rank}(P(AQ)) = \operatorname{rank}(AQ) = \operatorname{rank} A$ ((b), (c))

□

Theorem 16. Let $A \in \mathbb{R}^{m\times n}$.

(a) $\mathcal{N}(A'A) = \mathcal{N}(A)$.

(b) $\operatorname{rank}(A'A) = \operatorname{rank} A = \operatorname{rank}(AA')$.

(c) $\mathcal{C}(A'A) = \mathcal{C}(A')$ and $\mathcal{C}(AA') = \mathcal{C}(A)$.

Proof.

(a) Lecture 1, Theorem 8:

$Ax = 0 \implies A'Ax = A'0 = 0$
$A'Ax = 0 \implies (Ax)'(Ax) = x'A'Ax = 0$, the sum of the squares of the entries of $Ax \implies Ax = 0$

(b) $A'A$ and A have n columns; Theorem 11:

$\operatorname{rank}(A'A) = n - \dim \mathcal{N}(A'A) = n - \dim \mathcal{N}(A) = \operatorname{rank} A$ ((a))
A' in place of A : $\operatorname{rank}(AA') = \operatorname{rank} A' = \operatorname{rank} A$ ($(A')' = A$; Theorem 12(a))

(c) Theorems 15(a), 5(a):

$\mathcal{C}(A'A) \subseteq \mathcal{C}(A')$, both of dimension $\operatorname{rank} A$ ((b), Theorem 12(a)) $\implies \mathcal{C}(A'A) = \mathcal{C}(A')$
$\mathcal{C}(AA') \subseteq \mathcal{C}(A)$, both of dimension $\operatorname{rank} A$ ((b)) $\implies \mathcal{C}(AA') = \mathcal{C}(A)$

□

So $A'A \in \mathbb{R}^{n\times n}$ is invertible $\iff \operatorname{rank} A = n$ (Theorems 16(b), 12(d)).

Example 17. $A = \operatorname{diag}(1,1,1,0)$ and $B = \operatorname{diag}(0,0,1,1)$ in $\mathbb{R}^{4\times 4}$: find $\operatorname{rank} A$, $\operatorname{rank} B$, $\operatorname{rank}(AB)$ and $\operatorname{rank}(A'A)$.

Solution.

A zero column adds nothing to the span (Lecture 3, Theorem 9(c)); unit vectors are independent (Lecture 3, Definition 14, Proposition 12(b)):

$A = [e_1\ e_2\ e_3\ 0]$: $\mathcal{C}(A) = \operatorname{span}\{e_1, e_2, e_3\}$, $\operatorname{rank} A = 3$
$B = [0\ 0\ e_3\ e_4]$: $\mathcal{C}(B) = \operatorname{span}\{e_3, e_4\}$, $\operatorname{rank} B = 2$
$\operatorname{row}_i(AB) = a_{ii} \operatorname{row}_i(B)$ (Lecture 1, Theorem 4(c))
$AB = \operatorname{diag}(0,0,1,0) = [0\ 0\ e_3\ 0]$: $\operatorname{rank}(AB) = 1 < \min\{3,2\}$ (Theorem 15(a))
$A' = A$ (diagonal), $A^2 = A$ ($\operatorname{row}_i(A^2) = a_{ii} \operatorname{row}_i(A)$) (Lecture 1, Definition 7)
$A'A = A^2 = A$: $\operatorname{rank}(A'A) = 3 = \operatorname{rank} A$ (Theorem 16(b))

Practice. The Exercises and Problems of the textbook, Chapter 4: bases and dimension, the four subspaces, rank, consistency of $Ax = b$, and the rank of products and of $A'A$.